CS 411W Professional Workforce Dev II

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09/01/2026

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Version: 1.1

Lab 1: Project Outline

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1. Introduction

Internet users are subjected to rising levels of misinformation from online sources and platforms. Misinformation spreads rapidly across social media, news sites, and online communities, a problem that is heavily exacerbated by the rise of AI generated content. The scale of the issue is vast, affecting everyday web browsing and social media usage. Approximately 64% of U.S. adults report that misinformation causes major confusion. In addition to this, while roughly 94% of adults say they check the accuracy of their news sources, not all do so before sharing information with others. Consequently, an overwhelming 97% of surveyed individuals want stricter measures to limit the spread of misinformation.

The current process for combating this issue is often slow, confusing, and time-consuming. Manual validation requires extensive online searches and cross-referencing of sources, placing a heavy burden on the user. Existing content verification features offered by specific websites and platforms are limited in their scope and accessibility. These tools frequently suffer from platform exclusivity, popularity bias, and a primary focus on basic text verification. Current AI and chatbot offerings also fail to provide an intuitive, reliable, all-in-one solution. Additionally, manual verification is easily outpaced by the sheer volume of fake content that is being generated by AI tools.

Ultimately, users lack a universal and on-demand method for quickly verifying the information they encounter on the internet, and the burden of manual validating content results in many users skipping the process entirely. The required solution must be automated, fast, and capable of providing clear, evidence-based credibility assessments. Therefore, we are proposing FAICT - the Fact AI Checking Tool, to help fill in these gaps and make browsing the web a more reliable experience across all web-based content.

2. Product Description

FAICT is a comprehensive software tool designed to verify information in real-time across text, audio, and video formats. The system works by automatically extracting claims from user-provided content and analyzing them against reliable, trusted sources. The primary goal of FAICT is to provide clear credibility assessments accompanied by cited evidence and to detect AI-generated content. By automating and accelerating the verification process, FAICT's core objective is to help internet users quickly identify misinformation, reduce blind trust in unverified claims, and empower them to make better informed decisions.

2.1. Key Product Features and Capabilities

2.1.1. **Cross-Platform Fact-Checking**: FAICT allows users to verify information seamlessly from a vast amount of online sources.

2.1.2. **Real-Time Audio Verification**: The tool features a speech-to-text module, enabling live verification for podcasts, videos, and real-time conversations.

2.1.3. **Video Processing:** FAICT can convert video uploads into transcripts to facilitate automated fact-checking.

2.1.4. **AI Content Detection**: The software includes an AI detection module to identify AI-generated content across various forms of media.

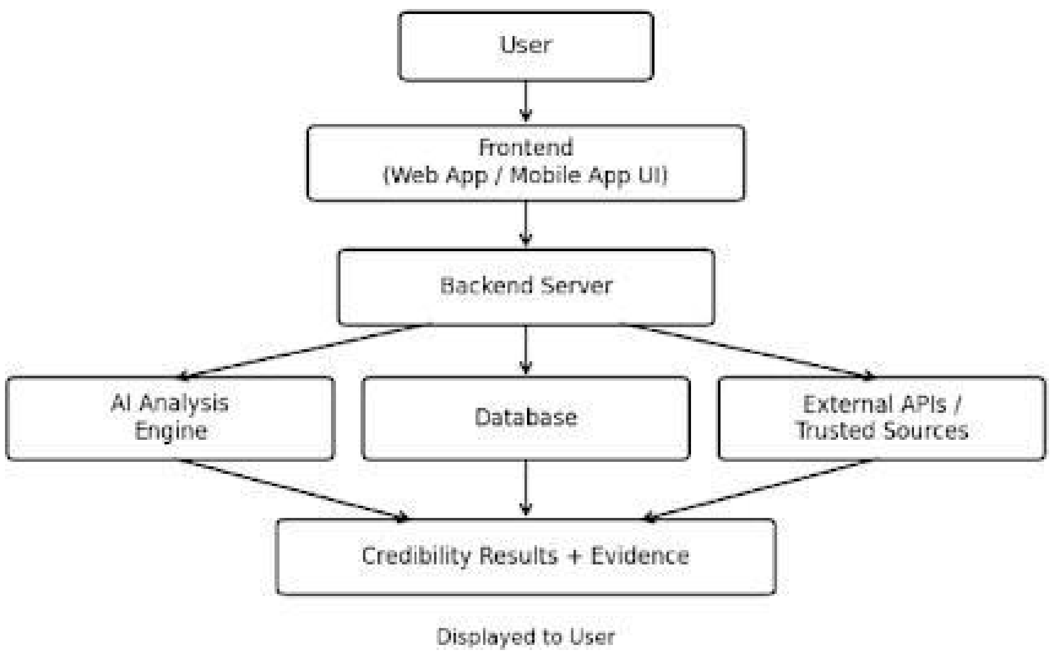
2.1.5. **Credibility Scoring**: FAICT provides clear credibility ratings - such as true, misleading, false, or uncertain - supplemented with direct explanations and trusted sources.

2.1.6. **Speed and Accessibility**: The software automates the fact-checking process while remaining easily accessible via a browser extension or mobile app.

2.2. Major Components (Hardware/Software)

FAICT is a software-based solution accessible via the internet. The software is being developed using a Python backend, HTML/CSS/JavaScript frontend interface, and a PostgreSQL or MongoDB database. As outlined in the Major Functional Components Diagram, the architecture is structured as follows:

Figure 1: Major Functional Components Diagram



2.2.1. **Frontend Interface**: A web app and mobile app UI that handles user inputs - from a variety of media forms - and displays the results dashboard.

2.2.2. **Backend Server**: Acts as the central integration point connecting the frontend GUI to the database, AI analysis engine, and external APIs.

2.2.3. **AI Analysis Engine**: Responsible for processing claim analysis and classification to determine validity of the submitted information.

2.2.4. **Processing Modules**: Includes specialized subsystems for speech-to-text conversion, video-to-transcript processing, and AI generation detection.

2.2.5. **Database**: Securely stores user data, submission records, structured claims, and analysis results.

2.2.6. **External APIs**: Connects the system to external web searches and trusted fact sources to collect supporting evidence.

3. Identification of Case Study

FAICT is primarily being developed for industry professionals who rely on real-time content verification to perform their jobs safely and efficiently. This target customer base includes journalists, content creators, and online moderators. Additionally, the product is built for organizations that enforce strict validity requirements, such as academic institutions, research facilities, and risk management teams. These groups require a tool like FAICT because manual validation costs them significant time and leaves them vulnerable to the rapid spread of AI-generated misinformation. Looking forward, FAICT seeks to expand our user base to everyday internet users who need a fast, cross-platform method to verify the increasing volume of questionable content they encounter during casual web browsing and social media use.

4. Glossary

4.1. AI – Artificial Intelligence

4.2. API – Application Programming Interface

4.3. FAICT – the Fact AI Checking Tool

4.4. GUI – Graphical User Interface

4.5. IDE – Integrated Development Environment

4.6. LLM – Large Language Model

4.7. AI-Generated Content – Text, images, audio, or video created or heavily modified by artificial intelligence tools.

4.8. Browser Extension – A small software add-on that adds features or functionality to a web browser.

4.9. Credibility Score – A rating that estimates how trustworthy or reliable a claim appears based on supporting evidence.

4.10. Disinformation – False or misleading information that is intentionally created or shared to deceive people.

4.11. Fact-Checking – The process of verifying whether a claim is accurate by comparing it against reliable sources.

4.12. Misinformation – False or misleading information that may be shared regardless of whether there is intent to deceive.

4.13. Speech-to-Text – Technology that converts spoken audio into written text.

4.14. Trusted Sources – Reliable references, databases, publications, or organizations used to verify claims.

4.15. **Verification** – The process of confirming whether information is accurate, reliable, or supported by evidence.

4.16. **Video-to-Transcript** – The process of converting spoken content from a video into written text for analysis.

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